SPECULA • DIGITAL FORENSIC EXAMINATION

Digital Forensic Examination Report

COURT SUBMISSION DRAFT — EXAMINER REVIEW REQUIRED

Case LOCAL-2f783c8d4d5941c2a6ad4442db1a6729

1. Mandate, scope and document control

Field	Recorded value
Case identifier	LOCAL-2f783c8d4d5941c2a6ad4442db1a6729
Declared matter/title	Not recorded
Declared examiner	Not recorded
Declared organization	Not recorded
Declared examination authority	Not recorded
Investigation start (UTC)	2026-10-27T08:56:23.264192+00:00
Document generated (UTC)	2026-10-27T09:08:31.772730+00:00
Evidence completeness	incomplete
Normalized snapshot SHA-256	7f35242216b1784335478394d838b60ec905734741ff2ea440aa82a1ee18d659

Scope is limited to the preserved records linked to this case. Supplied metadata is declarative. No conclusion about activity outside this collection window or unsupported log types is made.

2. Executive examination result

20 normalized evidence records and 28 evidence-cited detection findings were available. An absence of supported detections does not establish that a device is uncompromised.

Measure	Value
Case state	hitl_review
Acceptance status	incomplete
Earliest recorded event	2025-01-01T00:00:00+00:00

Measure	Value
Latest recorded event	2026-10-13T03:22:00+00:00
Executed agent steps	18

3. Evidence receipt, integrity and custody

Item	Recorded value
Collection source	uploaded_zip
Original archive SHA-256	d679313df39c4b4e6fdccef7685ed08fe259f84b8c2fcecce67b46af21320734
Archive byte length	5378
Receipt time (UTC)	2026-10-27T08:56:05.432087+00:00
Custody scope	Receipt recorded; acquisition and prior custody are uploader declarations

Original file	Bytes	SHA-256	Record count
edr/events.jsonl	909	594a2ae67003791958f420c05b382496d4824ef545d532334841dd9dd5fbb96c	3
gcp_audit/events.json	801	f3eae391e5fd30721920d6edd93996645b76f86f6938f61887e0c58bd7174477	2
malware/findings.json	535	99cf588b8e9d3966d9e20e58affadf6fee12d9545469b1cf92c961619f30e950	1
memory/findings.json	519	3d00a8ef5af1b292ae7cc34b4b92a0384c37ce3f78ade5d0249d644563e0a632	2
mft/records.json	614	a7155b6ecf08305f33c9d0399519c6a5efa7446ff0275025af1dc52b7e0fc899	1
ueba/events.json	751	aaca257c798307cfeebb973bc028ad67c283eabfc9e1a9f971ea204220d9099a	3
vuln_scan/findings.json	475	1f6c1c49b639933580d534135a0c00f1e8f383a685e8d62f069d758db08ddac4	2

Original file	Bytes	SHA-256	Record count
windows/Security.json	1606	7df5eaf5952ec760fa81a35eb4f2a94a2dc814db25227155dfb0ae5196e8642c	5
windows/Sysmon.jsonl	1696	17ea47ae3af61e0c1b2823d6acd490526484919eb156f8cfd7ca3a5c62560ef1	5

Original archive/file hashes identify bytes at receipt. Normalized record hashes identify derived data. They do not prove authenticity at acquisition, independent custody verification or legal admissibility. Hash verification on every subsequent transfer remains a reviewer responsibility.

Uploader-declared acquisition and custody


```
{
  "case_id": "SPECULA-TEST-001",
  "created_at": "2026-10-13T04:46:12.595124+00:00",
  "description": "Synthetic DFIR fixture for Specula pipeline validation. Contains intentional attack
indicators and anti-forensic tampering.",
  "examiner": "OpenAI synthetic fixture generator",
  "expected_attack_story": {
    "expected_tampering": [
      "Windows Security log clearing via wevtutil",
      "MFT timestamp inconsistency/timestomping",
      "malware executable staged under ProgramData"
    ],
    "expected_techniques": [
      "T1110",
      "T1078",
      "T1059.001",
      "T1003",
      "T1071.001",
      "T1041",
      "T1070.001",
      "T1070.006"
    ],
    "summary": "External credential attack -> successful remote logon -> PowerShell staging -> malware
drop/execution -> LSASS access -> C2/exfiltration -> log clearing -> timestomping; separate suspicious GCP
service-account activity."
  },
  "files": {
    "edr/events.jsonl": {
      "sha256": "594a2ae67003791958f420c05b382496d4824ef545d532334841dd9dd5fbb96c",
      "size_bytes": 909
    },
    "gcp_audit/events.json": {
      "sha256": "f3eae391e5fd30721920d6edd93996645b76f86f6938f61887e0c58bd7174477",
      "size_bytes": 801
    },
    "malware/findings.json": {
      "sha256": "99cf588b8e9d3966d9e20e58affadf6fee12d9545469b1cf92c961619f30e950",
      "size_bytes": 535
    },
```

```
    "memory/findings.json": {
      "sha256": "3d00a8ef5af1b292ae7cc34b4b92a0384c37ce3f78ade5d0249d644563e0a632",
      "size_bytes": 519
    },
    "mft/records.json": {
      "sha256": "a7155b6ecf08305f33c9d0399519c6a5efa7446ff0275025af1dc52b7e0fc899",
      "size_bytes": 614
    },
    "ueba/events.json": {
      "sha256": "aaca257c798307cfeebb973bc028ad67c283eabfc9e1a9f971ea204220d9099a",
      "size_bytes": 751
    },
    "vuln_scan/findings.json": {
      "sha256": "1f6c1c49b639933580d534135a0c00f1e8f383a685e8d62f069d758db08ddac4",
      "size_bytes": 475
    },
    "windows/Security.json": {
      "sha256": "7df5eaf5952ec760fa81a35eb4f2a94a2dc814db25227155dfb0ae5196e8642c",
      "size_bytes": 1606
    },
    "windows/Sysmon.jsonl": {
      "sha256": "17ea47ae3af61e0c1b2823d6acd490526484919eb156f8cfd7ca3a5c62560ef1",
      "size_bytes": 1696
    }
  },
  "synthetic": true
}
```


4. Methods and reproducibility

1. Preserve submitted bytes and inventory original files.
2. Apply security screening and supported source normalization to OCSF records with deterministic UIDs.
3. Read case-linked evidence; apply deterministic triage and supported detection rules.
4. Record model-assisted synthesis and debate as analysis requiring verification.
5. Render factual exhibits from saved structured records and retain incompleteness flags.

Runtime	Version
Python	3.10.22
Report template	specula-examination-v1

Source files measured at investigation start


```
{
  "packages": {
    "capstone": "5.0.9",
    "chromadb": "1.5.9",
    "duckdb": "1.5.6",
    "fastapi": "0.143.0",
    "langgraph": "1.2.15",
```

```
    "neo4j": "6.4.0",
    "pefile": "2024.8.26",
    "pydantic": "2.14.1",
    "volatility3": "2.28.2"
  },
  "python": "3.10.22",
  "source_sha256": {
    "src/agents/behavior_analysis.py": "aa3dcf364f58c1f1c80fb7af22ee1d80194988cae7c5f4cd15c9de82482504b7",
    "src/agents/cape_sandbox.py": "04ea25c7931f1f191c68e741f814a5ffcb4098cbe638873e6460ead930dc9f1f",
    "src/agents/case_analysis.py": "3366c41a13cdd04562b50a4529ad6fb9b18d18c0f3f358c3c744fae82d44c3f8",
    "src/agents/case_analytics.py": "c7c5f9a7d4012a1fcc9dfca7f9da182ff3d083b4a12c32e1b66c3b6d922a7f82",
    "src/agents/evidence_review.py": "c4bdc30dbd1d7753a3c5e43ba85a43b36dcfc3a8de4a7480f591188506c9ead7",
    "src/agents/examination_reports.py": "049a670631b7b51fd516a27b7fb7403892037d2deca3111e12807880a4d842bc",
    "src/agents/exploit_paths.py": "db82ef39538cc07346b52cc5deed83e0cc723dd209587aaec1438bb261464a30",
    "src/agents/forensic_tools.py": "3b29c3c44f8db3c50a63034cd64a753b4e9bde9bc7f8cca6388861d8c8e17a68",
    "src/agents/grounded_report.py": "022066697fd4a634ab7358baf5895f502eea747904fb632b95d582ffa702b327",
    "src/agents/log_analysis/detection_rules.py":
"75e9c30a1064af27323b5c47be3d6bb4849b982c47321b90b572aa625828ee2f",
    "src/agents/log_analysis_factory.py":
"74063c9ff371db5683738587562ea98fa6ddeb9892a0abb5e57236704718fa28",
    "src/agents/network_specialist.py": "999b617baa232380ff232145abc0c6d48a0a61fb47bf3717a41c1962dae2c326",
    "src/agents/nodes.py": "4ccc3404fe0b176b8c3cb31abf295ffeca1fb2ccccbd718217fc17cce4289e5f",
    "src/agents/ollama_backend.py": "6da4732d767c6fdbe3069741a1def89193e320445eb192a367f8acb1fc61149d",
    "src/graph/process_lineage.py": "d73bc50bd82c83f5f17437c12a51d541230d0fcbcd029433cc7d47c8530f33ad",
    "src/ingestion/indexing/case_evidence.py":
"40e5f35ead1b5a234c162e81cea370946db4a39e48ae084250fd8501951705bb",
    "src/ingestion/run_pipeline.py": "468752e6665d6e68ab29a25b80bc1f5944f27aa6f0fb840042647b5675b0d9cb",
    "src/schemas/ocsf_base.py": "6235c5c3d2a496602e51723b23032f498a9767d152c87cafdbedff18a2660b25"
  }
}
```


Current report-rendering environment


```
{
  "packages": {
    "capstone": "5.0.9",
    "chromadb": "1.5.9",
    "duckdb": "1.5.6",
    "fastapi": "0.143.0",
    "langgraph": "1.2.15",
    "neo4j": "6.4.0",
    "pefile": "2024.8.26",
    "pydantic": "2.14.1",
    "volatility3": "2.28.2"
  },
  "python": "3.10.22",
  "source_sha256": {
    "src/agents/behavior_analysis.py": "aa3dcf364f58c1f1c80fb7af22ee1d80194988cae7c5f4cd15c9de82482504b7",
    "src/agents/cape_sandbox.py": "04ea25c7931f1f191c68e741f814a5ffcb4098cbe638873e6460ead930dc9f1f",
    "src/agents/case_analysis.py": "3366c41a13cdd04562b50a4529ad6fb9b18d18c0f3f358c3c744fae82d44c3f8",
    "src/agents/case_analytics.py": "c7c5f9a7d4012a1fcc9dfca7f9da182ff3d083b4a12c32e1b66c3b6d922a7f82",
    "src/agents/evidence_review.py": "c4bdc30dbd1d7753a3c5e43ba85a43b36dcfc3a8de4a7480f591188506c9ead7",
    "src/agents/examination_reports.py": "049a670631b7b51fd516a27b7fb7403892037d2deca3111e12807880a4d842bc",
    "src/agents/exploit_paths.py": "db82ef39538cc07346b52cc5deed83e0cc723dd209587aaec1438bb261464a30",
    "src/agents/forensic_tools.py": "3b29c3c44f8db3c50a63034cd64a753b4e9bde9bc7f8cca6388861d8c8e17a68",
    "src/agents/grounded_report.py": "022066697fd4a634ab7358baf5895f502eea747904fb632b95d582ffa702b327",
    "src/agents/log_analysis/detection_rules.py":
"75e9c30a1064af27323b5c47be3d6bb4849b982c47321b90b572aa625828ee2f",
```

```
    "src/agents/log_analysis_factory.py":
"74063c9ff371db5683738587562ea98fa6ddeb9892a0abb5e57236704718fa28",
    "src/agents/network_specialist.py": "999b617baa232380ff232145abc0c6d48a0a61fb47bf3717a41c1962dae2c326",
    "src/agents/nodes.py": "4ccc3404fe0b176b8c3cb31abf295ffeca1fb2ccccbd718217fc17cce4289e5f",
    "src/agents/ollama_backend.py": "6da4732d767c6fdbe3069741a1def89193e320445eb192a367f8acb1fc61149d",
    "src/graph/process_lineage.py": "d73bc50bd82c83f5f17437c12a51d541230d0fcbcd029433cc7d47c8530f33ad",
    "src/ingestion/indexing/case_evidence.py":
"40e5f35ead1b5a234c162e81cea370946db4a39e48ae084250fd8501951705bb",
    "src/ingestion/run_pipeline.py": "468752e6665d6e68ab29a25b80bc1f5944f27aa6f0fb840042647b5675b0d9cb",
    "src/schemas/ocsf_base.py": "6235c5c3d2a496602e51723b23032f498a9767d152c87cafdbedff18a2660b25"
  }
}
```


Current source fingerprints do not prove which source version executed a historical investigation.

Timestamp conversion does not independently verify source clocks. Local evidence hashing and storage do not establish Hyperledger Fabric anchoring. Model outputs are not admitted as measured evidence.

5. Rule 702 and Daubert review record

Submission readiness: **NOT CERTIFIED — HUMAN REVIEW REQUIRED**. This automated draft is not an expert qualification, attestation or judicial admissibility finding.

Review factor	Current evidence / gap
Qualified expert and helpfulness (Rule 702)	Examiner qualifications, relevance and opinion scope require documented expert review.
Sufficient facts and reliable application (Rule 702)	Case exhibits and limitations are listed; sufficiency and reliable application require review.
Testability and validation	Reproducible component fingerprints included; case-specific independent validation evidence not attached.
Peer review / publication	No independent peer-review evidence for this complete system recorded.
Known or potential error rate	No validated system sensitivity, specificity, false-positive or false-negative rate recorded.
Operational standards	OCSF mapping and deterministic IDs used; laboratory accreditation and SOP compliance not established.
General acceptance	Not established by use of SHA-256, OCSF or an LLM alone.
Custody and authenticity	Receipt hashes are documented when available; earlier acquisition/custody declarations require verification.

Authorities: Federal Rules of Evidence, Rule 702 (December 1, 2025); Daubert v. Merrell Dow Pharmaceuticals, 509 U.S. 579 (1993). Applicable jurisdiction and evidentiary procedure must be

confirmed by counsel.

6. Additional evidence and follow-up examinations

Unanswered acquisition requests remain limitations; missing artifacts are not negative findings. Targeted memory regions or process minidumps can be supplied without requiring a whole-RAM image.


```
[
  {
    "acquisition_policy": "Add preserved case evidence with source time, host, digest and acquisition
provenance; never execute samples locally.",
    "artifact_type": "authentication_and_privilege_audit",
    "case_id": "LOCAL-2f783c8d4d5941c2a6ad4442db1a6729",
    "dfkg_refs": [
      "ef180ee7f7b9b03e282ce8cdadc2f02041c35a1a06ae3aab2a40a898f70a5974"
    ],
    "hosts": [
      "host-WS-17"
    ],
    "id": "1fd20a86067bde289541ed8d295698c9a84b06d01907b09876d6da455a313909",
    "question": "Investigate the process tree and memory dumps of svchost-helper.exe and lsass.exe for signs
of credential extraction.",
    "reason": "Available sources cannot independently answer this question.",
    "recorded_times": [
      "2026-10-13T03:13:00Z"
    ],
    "requested_by": "identity",
    "status": "requested"
  },
  {
    "acquisition_policy": "Add preserved case evidence with source time, host, digest and acquisition
provenance; never execute samples locally.",
    "artifact_type": "targeted_memory_region_or_process_minidump",
    "case_id": "LOCAL-2f783c8d4d5941c2a6ad4442db1a6729",
    "dfkg_refs": [
      "12c4270ebde02352981786b45f267276732f64fee9fc8de75cd8b86b0b215f06",
      "d8f86bc22440cafcde1591af767b093bc7e90288cfbca010bf92055d4a093b64"
    ],
    "hosts": [
      "host-WS-17"
    ],
    "id": "383bb879d0c6f11bcc9c847064829422ad1c54fab71a44402e42b2d00211b755",
    "question": "Independently examine the underlying artifact for these exported findings.",
    "reason": "Available sources cannot independently answer this question.",
    "recorded_times": [
      "2026-10-13T03:13:00Z"
    ],
    "requested_by": "memory_forensics",
    "status": "requested"
  },
  {
    "acquisition_policy": "Add preserved case evidence with source time, host, digest and acquisition
provenance; never execute samples locally.",
    "artifact_type": "historical_user_activity",
```

```
    "case_id": "LOCAL-2f783c8d4d5941c2a6ad4442db1a6729",
    "dfkg_refs": [
      "9306da2738863b2503ec39fd2d814104fcaf96e857c18d4ddda967ff4928a1a8"
    ],
    "hosts": [
      "host-WS-17"
    ],
    "id": "6365c0c87c19a98088b3148028775263f0da4e503ebd810993ab468f95048c4f",
    "question": "Investigate the source IP (203.0.113.45) and verify if it is associated with known
malicious activity. Check for any additional logs or network traffic related to the administrator user.",
    "reason": "Available sources cannot independently answer this question.",
    "recorded_times": [
      "2026-10-13T03:07:00Z"
    ],
    "requested_by": "insider_threat",
    "status": "requested"
  },
  {
    "acquisition_policy": "Add preserved case evidence with source time, host, digest and acquisition
provenance; never execute samples locally.",
    "artifact_type": "native_pe_sample",
    "case_id": "LOCAL-2f783c8d4d5941c2a6ad4442db1a6729",
    "dfkg_refs": [
      "7798710c4e2200c1c09ce0a586d759092692b9d46e057922d0f78adeb10cf379"
    ],
    "hosts": [
      "host-WS-17"
    ],
    "id": "64508034d528f6b62a991b9d42c4e7111e153297ebaa2728bbde1a79b6afb54c",
    "question": "Request additional analysis or examination of the executable to determine its nature and
intent.",
    "reason": "Available sources cannot independently answer this question.",
    "recorded_times": [
      "2026-10-13T03:12:00Z"
    ],
    "requested_by": "malware_stylometry",
    "status": "requested"
  },
  {
    "acquisition_policy": "Add preserved case evidence with source time, host, digest and acquisition
provenance; never execute samples locally.",
    "artifact_type": "native_pe_sample",
    "case_id": "LOCAL-2f783c8d4d5941c2a6ad4442db1a6729",
    "dfkg_refs": [
      "49d1830b659fd7f78ec440497974cca6cd7d2245de23129649949858743f8e48"
    ],
    "hosts": [
      "host-WS-17"
    ],
    "id": "9cc848d01403b33942a1c8175dc0a66013c9ebcb03df6afc13893da3da5e8ab3",
    "question": "Request additional analysis or examination of the executable to determine its nature and
intent.",
    "reason": "Available sources cannot independently answer this question.",
    "recorded_times": [
      "2026-10-13T03:12:00Z"
    ],
    "requested_by": "malware_stylometry",
    "status": "requested"
  },
```

```
  {
    "acquisition_policy": "Add preserved case evidence with source time, host, digest and acquisition
provenance; never execute samples locally.",
    "artifact_type": "authentication_and_privilege_audit",
    "case_id": "LOCAL-2f783c8d4d5941c2a6ad4442db1a6729",
    "dfkg_refs": [
      "1890fa9e2c4b9d7700bf83f3568cf731f64e336459e0a8b9cdf541ff0b0a13bb",
      "7d0fe397f89de4360afd58e97dd9da02a9d526f77fe44db22232e2c4bc0205e0",
      "99a5c5a73b882995178dcc5c9dc757d03c3e2cacc3d53ff142158bb5bfec7198"
    ],
    "hosts": [
      "host-WS-17"
    ],
    "id": "a23d3acd9ac0db95ca2f9a8dfdfa27975978ab916767038e29d01761d54a0e7e",
    "question": "Investigate the source IP for any signs of automated attacks or repeated attempts to access
the system.",
    "reason": "Available sources cannot independently answer this question.",
    "recorded_times": [
      "2026-10-13T03:05:00Z",
      "2026-10-13T03:06:00Z",
      "2026-10-13T03:07:00Z"
    ],
    "requested_by": "identity",
    "status": "requested"
  },
  {
    "acquisition_policy": "Add preserved case evidence with source time, host, digest and acquisition
provenance; never execute samples locally.",
    "artifact_type": "historical_user_activity",
    "case_id": "LOCAL-2f783c8d4d5941c2a6ad4442db1a6729",
    "dfkg_refs": [
      "e29aaaf03ab7cb8d40ca54ad8a6966642fd2b1fe446564e84166f01f2bc132ad"
    ],
    "hosts": [
      "host-WS-17"
    ],
    "id": "a69119473558106dc58220ecf780909a25445b510c64b44c85d518e6d69b8be9",
    "question": "Investigate the destination IP (198.51.100.77) and verify if it is associated with known
malicious activity. Check for any additional logs or network traffic related to the administrator user.",
    "reason": "Available sources cannot independently answer this question.",
    "recorded_times": [
      "2026-10-13T03:14:00Z"
    ],
    "requested_by": "insider_threat",
    "status": "requested"
  },
  {
    "acquisition_policy": "Add preserved case evidence with source time, host, digest and acquisition
provenance; never execute samples locally.",
    "artifact_type": "authentication_and_privilege_audit",
    "case_id": "LOCAL-2f783c8d4d5941c2a6ad4442db1a6729",
    "dfkg_refs": [
      "84627e88414ba5159935eb8979a6f2f8542beea2c39dc765a8b40b716850be57"
    ],
    "hosts": [
      "host-WS-17"
    ],
    "id": "a8a9cd1a3f59b23ee90c742b4003077e2ca7251a8faec54afc38aac319d634d5",
    "question": "Review the privilege assignment logs and correlate them with other events to assess
```

```
potential privilege escalation.",
    "reason": "Available sources cannot independently answer this question.",
    "recorded_times": [
      "2026-10-13T03:11:00Z"
    ],
    "requested_by": "identity",
    "status": "requested"
  },
  {
    "acquisition_policy": "Add preserved case evidence with source time, host, digest and acquisition
provenance; never execute samples locally.",
    "artifact_type": "targeted_memory_region_or_process_minidump",
    "case_id": "LOCAL-2f783c8d4d5941c2a6ad4442db1a6729",
    "dfkg_refs": [
      "12c4270ebde02352981786b45f267276732f64fee9fc8de75cd8b86b0b215f06"
    ],
    "hosts": [
      "host-WS-17"
    ],
    "id": "cd2be55f13d03a514cb83f9908936771e0ed53e1bfe90f1b6c6b2c9ad83e4ac3",
    "question": "Verify the synthetic nature of the finding and cross-check with other memory artifacts or
logs for potential credential leaks.",
    "reason": "Available sources cannot independently answer this question.",
    "recorded_times": [
      "2026-10-13T03:13:00Z"
    ],
    "requested_by": "memory_forensics",
    "status": "requested"
  },
  {
    "acquisition_policy": "Add preserved case evidence with source time, host, digest and acquisition
provenance; never execute samples locally.",
    "artifact_type": "targeted_memory_region_or_process_minidump",
    "case_id": "LOCAL-2f783c8d4d5941c2a6ad4442db1a6729",
    "dfkg_refs": [
      "d8f86bc22440cafcde1591af767b093bc7e90288cfbca010bf92055d4a093b64"
    ],
    "hosts": [
      "host-WS-17"
    ],
    "id": "cfa1941edc2e539429ad6dda727bc156e5630460dd1f616849e7e7e77611e390",
    "question": "Perform a detailed analysis of the memory region using forensic tools to determine its
origin and potential malicious intent.",
    "reason": "Available sources cannot independently answer this question.",
    "recorded_times": [
      "2026-10-13T03:13:00Z"
    ],
    "requested_by": "memory_forensics",
    "status": "requested"
  }
]
```


Examination receipts


```
[
  {
    "anchor_refs": [
      "7798710c4e2200c1c09ce0a586d759092692b9d46e057922d0f78adeb10cf379"
```

```
    ],
    "basis": [
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        "canonical_host_id": "host-WS-17",
        "causal": false,
        "keys": [
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        ],
        "source": "7798710c4e2200c1c09ce0a586d759092692b9d46e057922d0f78adeb10cf379",
        "target": "12c4270ebde02352981786b45f267276732f64fee9fc8de75cd8b86b0b215f06"
      },
      {
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        "causal": false,
        "keys": [
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        ],
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        "target": "1890fa9e2c4b9d7700bf83f3568cf731f64e336459e0a8b9cdf541ff0b0a13bb"
      },
      {
        "canonical_host_id": "host-WS-17",
        "causal": false,
        "keys": [
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        ],
        "source": "7798710c4e2200c1c09ce0a586d759092692b9d46e057922d0f78adeb10cf379",
        "target": "450bb3a062a93585350e3e89e9ba6c53ff2dc75ffa3ca6908a732f9c53628bce"
      },
      {
        "canonical_host_id": "host-WS-17",
        "causal": false,
        "keys": [
          "source_process_uid_or_parent_uid"
        ],
        "source": "7798710c4e2200c1c09ce0a586d759092692b9d46e057922d0f78adeb10cf379",
        "target": "49d1830b659fd7f78ec440497974cca6cd7d2245de23129649949858743f8e48"
      },
      {
        "canonical_host_id": "host-WS-17",
        "causal": false,
        "keys": [
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        ],
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        "target": "7d0fe397f89de4360afd58e97dd9da02a9d526f77fe44db22232e2c4bc0205e0"
      },
      {
        "canonical_host_id": "host-WS-17",
        "causal": false,
        "keys": [
          "source_process_uid_or_parent_uid"
        ],
        "source": "7798710c4e2200c1c09ce0a586d759092692b9d46e057922d0f78adeb10cf379",
        "target": "84627e88414ba5159935eb8979a6f2f8542beea2c39dc765a8b40b716850be57"
      },
      {
        "canonical_host_id": "host-WS-17",
        "causal": false,
```

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      "keys": [
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      ],
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      "target": "9306da2738863b2503ec39fd2d814104fcaf96e857c18d4ddda967ff4928a1a8"
    },
    {
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      "causal": false,
      "keys": [
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      ],
      "source": "7798710c4e2200c1c09ce0a586d759092692b9d46e057922d0f78adeb10cf379",
      "target": "99a5c5a73b882995178dcc5c9dc757d03c3e2cacc3d53ff142158bb5bfec7198"
    },
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      "causal": false,
      "keys": [
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      ],
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      "target": "bd5eef55cd225d45b87b64b1db021c03c64ad4ff4933a2c7c8b3a003230e8b52"
    },
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      "canonical_host_id": "host-WS-17",
      "causal": false,
      "keys": [
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      ],
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      "target": "d8f86bc22440cafcde1591af767b093bc7e90288cfbca010bf92055d4a093b64"
    },
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      "causal": false,
      "keys": [
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      ],
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      "target": "e29aaaf03ab7cb8d40ca54ad8a6966642fd2b1fe446564e84166f01f2bc132ad"
    },
    {
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      "causal": false,
      "keys": [
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      ],
      "source": "7798710c4e2200c1c09ce0a586d759092692b9d46e057922d0f78adeb10cf379",
      "target": "ef180ee7f7b9b03e282ce8cdadc2f02041c35a1a06ae3aab2a40a898f70a5974"
    },
    {
      "canonical_host_id": "host-WS-17",
      "causal": false,
      "keys": [
        "source_process_uid_or_parent_uid"
      ],
      "source": "7798710c4e2200c1c09ce0a586d759092692b9d46e057922d0f78adeb10cf379",
      "target": "fb5ca28e786e848f56cbc1cc10f319615ef7b0a1abe142d5245003ab0ca0e66e"
```

```
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    ],
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      "12c4270ebde02352981786b45f267276732f64fee9fc8de75cd8b86b0b215f06",
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      "450bb3a062a93585350e3e89e9ba6c53ff2dc75ffa3ca6908a732f9c53628bce",
      "49d1830b659fd7f78ec440497974cca6cd7d2245de23129649949858743f8e48",
      "7798710c4e2200c1c09ce0a586d759092692b9d46e057922d0f78adeb10cf379",
      "7d0fe397f89de4360afd58e97dd9da02a9d526f77fe44db22232e2c4bc0205e0",
      "84627e88414ba5159935eb8979a6f2f8542beea2c39dc765a8b40b716850be57",
      "9306da2738863b2503ec39fd2d814104fcaf96e857c18d4ddda967ff4928a1a8",
      "99a5c5a73b882995178dcc5c9dc757d03c3e2cacc3d53ff142158bb5bfec7198",
      "bd5eef55cd225d45b87b64b1db021c03c64ad4ff4933a2c7c8b3a003230e8b52",
      "d8f86bc22440cafcde1591af767b093bc7e90288cfbca010bf92055d4a093b64",
      "e29aaaf03ab7cb8d40ca54ad8a6966642fd2b1fe446564e84166f01f2bc132ad",
      "ef180ee7f7b9b03e282ce8cdadc2f02041c35a1a06ae3aab2a40a898f70a5974",
      "fb5ca28e786e848f56cbc1cc10f319615ef7b0a1abe142d5245003ab0ca0e66e"
    ],
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relationships do not establish causation or authorization.",
    "matched_count": 14,
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        "uid": "12c4270ebde02352981786b45f267276732f64fee9fc8de75cd8b86b0b215f06"
      },
      {
        "outcome": "failed",
        "uid": "1890fa9e2c4b9d7700bf83f3568cf731f64e336459e0a8b9cdf541ff0b0a13bb"
      },
      {
        "outcome": "unknown",
        "uid": "450bb3a062a93585350e3e89e9ba6c53ff2dc75ffa3ca6908a732f9c53628bce"
      },
      {
        "outcome": "unknown",
        "uid": "49d1830b659fd7f78ec440497974cca6cd7d2245de23129649949858743f8e48"
      },
      {
        "outcome": "unknown",
        "uid": "7798710c4e2200c1c09ce0a586d759092692b9d46e057922d0f78adeb10cf379"
      },
      {
        "outcome": "failed",
        "uid": "7d0fe397f89de4360afd58e97dd9da02a9d526f77fe44db22232e2c4bc0205e0"
      },
      {
        "outcome": "unknown",
        "uid": "84627e88414ba5159935eb8979a6f2f8542beea2c39dc765a8b40b716850be57"
      },
      {
        "outcome": "unknown",
        "uid": "9306da2738863b2503ec39fd2d814104fcaf96e857c18d4ddda967ff4928a1a8"
      },
      {
        "outcome": "confirmed",
        "uid": "99a5c5a73b882995178dcc5c9dc757d03c3e2cacc3d53ff142158bb5bfec7198"
      },
```

```
      {
        "outcome": "confirmed",
        "uid": "bd5eef55cd225d45b87b64b1db021c03c64ad4ff4933a2c7c8b3a003230e8b52"
      },
      {
        "outcome": "unknown",
        "uid": "d8f86bc22440cafcde1591af767b093bc7e90288cfbca010bf92055d4a093b64"
      },
      {
        "outcome": "unknown",
        "uid": "e29aaaf03ab7cb8d40ca54ad8a6966642fd2b1fe446564e84166f01f2bc132ad"
      },
      {
        "outcome": "unknown",
        "uid": "ef180ee7f7b9b03e282ce8cdadc2f02041c35a1a06ae3aab2a40a898f70a5974"
      },
      {
        "outcome": "unknown",
        "uid": "fb5ca28e786e848f56cbc1cc10f319615ef7b0a1abe142d5245003ab0ca0e66e"
      }
    ],
    "question": "Request additional analysis or examination of the executable to determine its nature and
intent.",
    "requested_by": "malware_stylometry",
    "returned_count": 14,
    "round": 1,
    "status": "complete",
    "task_id": "0bc95ad446e60823c25468453b1941855779712079329bd0055848826d68dbde",
    "tool": "process_lineage"
  },
  {
    "anchor_refs": [
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    ],
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    ],
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relationships do not establish causation or authorization.",
    "matched_count": 1,
    "outcomes": [
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        "outcome": "unknown",
        "uid": "9306da2738863b2503ec39fd2d814104fcaf96e857c18d4ddda967ff4928a1a8"
      }
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    "question": "Investigate the source IP (203.0.113.45) and verify if it is associated with known
malicious activity. Check for any additional logs or network traffic related to the administrator user.",
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    "returned_count": 1,
    "round": 1,
    "status": "complete",
    "task_id": "11f980e5f3950510054841736897d85d5a923735179fec7b365a471b791555e7",
    "tool": "related_records"
  },
  {
    "anchor_refs": [
```

```
      "e29aaaf03ab7cb8d40ca54ad8a6966642fd2b1fe446564e84166f01f2bc132ad"
    ],
    "basis": [],
    "cycle": 1,
    "dfkg_refs": [
      "e29aaaf03ab7cb8d40ca54ad8a6966642fd2b1fe446564e84166f01f2bc132ad"
    ],
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relationships do not establish causation or authorization.",
    "matched_count": 1,
    "outcomes": [
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        "outcome": "unknown",
        "uid": "e29aaaf03ab7cb8d40ca54ad8a6966642fd2b1fe446564e84166f01f2bc132ad"
      }
    ],
    "question": "Investigate the destination IP (198.51.100.77) and verify if it is associated with known
malicious activity. Check for any additional logs or network traffic related to the administrator user.",
    "requested_by": "insider_threat",
    "returned_count": 1,
    "round": 1,
    "status": "complete",
    "task_id": "24a334f525b5af77cf988fdc75c751fbc4b365a8b29aa80f8c0e6faf9e35146c",
    "tool": "related_records"
  },
  {
    "anchor_refs": [
      "ef180ee7f7b9b03e282ce8cdadc2f02041c35a1a06ae3aab2a40a898f70a5974"
    ],
    "basis": [],
    "cycle": 1,
    "dfkg_refs": [
      "ef180ee7f7b9b03e282ce8cdadc2f02041c35a1a06ae3aab2a40a898f70a5974"
    ],
    "limitation": "Case-scoped recorded identifiers only; absence is not proof of non-occurrence. Source
relationships do not establish causation or authorization.",
    "matched_count": 1,
    "outcomes": [
      {
        "outcome": "unknown",
        "uid": "ef180ee7f7b9b03e282ce8cdadc2f02041c35a1a06ae3aab2a40a898f70a5974"
      }
    ],
    "question": "Investigate the process tree and memory dumps of svchost-helper.exe and lsass.exe for signs
of credential extraction.",
    "requested_by": "identity",
    "returned_count": 1,
    "round": 1,
    "status": "complete",
    "task_id": "bf673f0504c0f9874ca1f5addf8bdb4019df86e4f93ae2268aa1d8f8d607d02f",
    "tool": "related_records"
  },
  {
    "anchor_refs": [
      "1890fa9e2c4b9d7700bf83f3568cf731f64e336459e0a8b9cdf541ff0b0a13bb",
      "7d0fe397f89de4360afd58e97dd9da02a9d526f77fe44db22232e2c4bc0205e0",
      "99a5c5a73b882995178dcc5c9dc757d03c3e2cacc3d53ff142158bb5bfec7198"
    ],
    "basis": [],
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```
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      "7d0fe397f89de4360afd58e97dd9da02a9d526f77fe44db22232e2c4bc0205e0",
      "99a5c5a73b882995178dcc5c9dc757d03c3e2cacc3d53ff142158bb5bfec7198"
    ],
    "limitation": "Case-scoped recorded identifiers only; absence is not proof of non-occurrence. Source
relationships do not establish causation or authorization.",
    "matched_count": 3,
    "outcomes": [
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        "outcome": "failed",
        "uid": "1890fa9e2c4b9d7700bf83f3568cf731f64e336459e0a8b9cdf541ff0b0a13bb"
      },
      {
        "outcome": "failed",
        "uid": "7d0fe397f89de4360afd58e97dd9da02a9d526f77fe44db22232e2c4bc0205e0"
      },
      {
        "outcome": "confirmed",
        "uid": "99a5c5a73b882995178dcc5c9dc757d03c3e2cacc3d53ff142158bb5bfec7198"
      }
    ],
    "question": "Investigate the source IP for any signs of automated attacks or repeated attempts to access
the system.",
    "requested_by": "identity",
    "returned_count": 3,
    "round": 1,
    "status": "complete",
    "task_id": "f8805d86d94279fa1d8b9d2215d438f6d5c8a3da8591d198dbde0ed2475dbb1c",
    "tool": "related_records"
  },
  {
    "anchor_refs": [
      "84627e88414ba5159935eb8979a6f2f8542beea2c39dc765a8b40b716850be57"
    ],
    "basis": [],
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    "dfkg_refs": [
      "84627e88414ba5159935eb8979a6f2f8542beea2c39dc765a8b40b716850be57"
    ],
    "limitation": "Case-scoped recorded identifiers only; absence is not proof of non-occurrence. Source
relationships do not establish causation or authorization.",
    "matched_count": 1,
    "outcomes": [
      {
        "outcome": "unknown",
        "uid": "84627e88414ba5159935eb8979a6f2f8542beea2c39dc765a8b40b716850be57"
      }
    ],
    "question": "Review the privilege assignment logs and correlate them with other events to assess
potential privilege escalation.",
    "requested_by": "identity",
    "returned_count": 1,
    "round": 1,
    "status": "complete",
    "task_id": "d6b3799bf50844ae34ffefec716a39f01c4fb07fcc5526212a2327467ab95c35",
    "tool": "related_records"
  },
```

```
  {
    "anchor_refs": [
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    ],
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        "canonical_host_id": "host-WS-17",
        "causal": false,
        "keys": [
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          "sha256"
        ],
        "source": "7798710c4e2200c1c09ce0a586d759092692b9d46e057922d0f78adeb10cf379",
        "target": "450bb3a062a93585350e3e89e9ba6c53ff2dc75ffa3ca6908a732f9c53628bce"
      },
      {
        "canonical_host_id": "host-WS-17",
        "causal": false,
        "keys": [
          "sha256"
        ],
        "source": "7798710c4e2200c1c09ce0a586d759092692b9d46e057922d0f78adeb10cf379",
        "target": "49d1830b659fd7f78ec440497974cca6cd7d2245de23129649949858743f8e48"
      },
      {
        "canonical_host_id": "host-WS-17",
        "causal": false,
        "keys": [
          "sha256"
        ],
        "source": "7798710c4e2200c1c09ce0a586d759092692b9d46e057922d0f78adeb10cf379",
        "target": "79a8f04efc3dac3410ed4d430ec0132309ec0a744503aa36e3666811cdd3ba42"
      }
    ],
    "cycle": 1,
    "dfkg_refs": [
      "450bb3a062a93585350e3e89e9ba6c53ff2dc75ffa3ca6908a732f9c53628bce",
      "49d1830b659fd7f78ec440497974cca6cd7d2245de23129649949858743f8e48",
      "7798710c4e2200c1c09ce0a586d759092692b9d46e057922d0f78adeb10cf379",
      "79a8f04efc3dac3410ed4d430ec0132309ec0a744503aa36e3666811cdd3ba42"
    ],
    "limitation": "Case-scoped recorded identifiers only; absence is not proof of non-occurrence. Source
relationships do not establish causation or authorization.",
    "matched_count": 4,
    "outcomes": [
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        "outcome": "unknown",
        "uid": "450bb3a062a93585350e3e89e9ba6c53ff2dc75ffa3ca6908a732f9c53628bce"
      },
      {
        "outcome": "unknown",
        "uid": "49d1830b659fd7f78ec440497974cca6cd7d2245de23129649949858743f8e48"
      },
      {
        "outcome": "unknown",
        "uid": "7798710c4e2200c1c09ce0a586d759092692b9d46e057922d0f78adeb10cf379"
      },
      {
        "outcome": "unknown",
```

```
        "uid": "79a8f04efc3dac3410ed4d430ec0132309ec0a744503aa36e3666811cdd3ba42"
      }
    ],
    "question": "Request additional analysis or examination of the executable to determine its nature and
intent.",
    "requested_by": "malware_stylometry",
    "returned_count": 4,
    "round": 2,
    "status": "complete",
    "task_id": "9d004603d98944e05b9c50c0b492436bd2739f2b5f2a900f6a8a0e4fce7d477c",
    "tool": "related_records"
  },
  {
    "anchor_refs": [
      "12c4270ebde02352981786b45f267276732f64fee9fc8de75cd8b86b0b215f06"
    ],
    "basis": [],
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    "dfkg_refs": [
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    ],
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relationships do not establish causation or authorization.",
    "matched_count": 1,
    "outcomes": [
      {
        "outcome": "unknown",
        "uid": "12c4270ebde02352981786b45f267276732f64fee9fc8de75cd8b86b0b215f06"
      }
    ],
    "question": "Verify the synthetic nature of the finding and cross-check with other memory artifacts or
logs for potential credential leaks.",
    "requested_by": "memory_forensics",
    "returned_count": 1,
    "round": 2,
    "status": "complete",
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    "tool": "related_records"
  },
  {
    "anchor_refs": [
      "49d1830b659fd7f78ec440497974cca6cd7d2245de23129649949858743f8e48"
    ],
    "basis": [
      {
        "canonical_host_id": "host-WS-17",
        "causal": false,
        "keys": [
          "sha256"
        ],
        "source": "49d1830b659fd7f78ec440497974cca6cd7d2245de23129649949858743f8e48",
        "target": "450bb3a062a93585350e3e89e9ba6c53ff2dc75ffa3ca6908a732f9c53628bce"
      },
      {
        "canonical_host_id": "host-WS-17",
        "causal": false,
        "keys": [
          "sha256"
        ],
```

```
        "source": "49d1830b659fd7f78ec440497974cca6cd7d2245de23129649949858743f8e48",
        "target": "7798710c4e2200c1c09ce0a586d759092692b9d46e057922d0f78adeb10cf379"
      },
      {
        "canonical_host_id": "host-WS-17",
        "causal": false,
        "keys": [
          "sha256"
        ],
        "source": "49d1830b659fd7f78ec440497974cca6cd7d2245de23129649949858743f8e48",
        "target": "79a8f04efc3dac3410ed4d430ec0132309ec0a744503aa36e3666811cdd3ba42"
      }
    ],
    "cycle": 1,
    "dfkg_refs": [
      "450bb3a062a93585350e3e89e9ba6c53ff2dc75ffa3ca6908a732f9c53628bce",
      "49d1830b659fd7f78ec440497974cca6cd7d2245de23129649949858743f8e48",
      "7798710c4e2200c1c09ce0a586d759092692b9d46e057922d0f78adeb10cf379",
      "79a8f04efc3dac3410ed4d430ec0132309ec0a744503aa36e3666811cdd3ba42"
    ],
    "limitation": "Case-scoped recorded identifiers only; absence is not proof of non-occurrence. Source
relationships do not establish causation or authorization.",
    "matched_count": 4,
    "outcomes": [
      {
        "outcome": "unknown",
        "uid": "450bb3a062a93585350e3e89e9ba6c53ff2dc75ffa3ca6908a732f9c53628bce"
      },
      {
        "outcome": "unknown",
        "uid": "49d1830b659fd7f78ec440497974cca6cd7d2245de23129649949858743f8e48"
      },
      {
        "outcome": "unknown",
        "uid": "7798710c4e2200c1c09ce0a586d759092692b9d46e057922d0f78adeb10cf379"
      },
      {
        "outcome": "unknown",
        "uid": "79a8f04efc3dac3410ed4d430ec0132309ec0a744503aa36e3666811cdd3ba42"
      }
    ],
    "question": "Request additional analysis or examination of the executable to determine its nature and
intent.",
    "requested_by": "log_analysis",
    "returned_count": 4,
    "round": 2,
    "status": "complete",
    "task_id": "6dece71d6cd69fcc20c38a79bf0a5c115f2e6140c1e71f2358293577b3bcc926",
    "tool": "related_records"
  },
  {
    "anchor_refs": [
      "d8f86bc22440cafcde1591af767b093bc7e90288cfbca010bf92055d4a093b64"
    ],
    "basis": [],
    "cycle": 1,
    "dfkg_refs": [
      "d8f86bc22440cafcde1591af767b093bc7e90288cfbca010bf92055d4a093b64"
    ],
```

```
    "limitation": "Case-scoped recorded identifiers only; absence is not proof of non-occurrence. Source
relationships do not establish causation or authorization.",
    "matched_count": 1,
    "outcomes": [
      {
        "outcome": "unknown",
        "uid": "d8f86bc22440cafcde1591af767b093bc7e90288cfbca010bf92055d4a093b64"
      }
    ],
    "question": "Perform a detailed analysis of the memory region using forensic tools to determine its
origin and potential malicious intent.",
    "requested_by": "memory_forensics",
    "returned_count": 1,
    "round": 2,
    "status": "complete",
    "task_id": "78fda9a07d1c44f8b116a6035c66bbeb33dc2f7d0f393a967d4bcb77cf68231a",
    "tool": "related_records"
  }
]
```


7. Recorded automated detection results

Automated screening results require examiner validation. Citations establish record membership, not independent correctness of an inference.

log_analysis

Deterministic unusual_parent_process signal: C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe; powershell.exe -NoProfile -ExecutionPolicy Bypass -EncodedCommand UwBZAE4AVABIAEUAVABJAEMAIAAtACAAdABlAHMAdAA=; user=administrator; source context={"event_id": 1}

Classification: unresolved; operation outcome: unknown

Evidence references: 098d0cfb012744a7a5c0fc785bda47b233f32165e4d3d8e809d05537d42a89e4

log_analysis

Command attempts Security-log clearing; outcome unknown.

Classification: suspicious; operation outcome: unknown

Evidence references: c0d8a9cb4a8f9cb43fba1fc83bd2fadb9c86afa6d16a15b400dc242185c71ba7

log_analysis

Executable launched from ProgramData: C:\ProgramData\svchost-helper.exe; location alone does not establish malware.

Classification: suspicious; operation outcome: unknown

Evidence references: 79a8f04efc3dac3410ed4d430ec0132309ec0a744503aa36e3666811cdd3ba42

log_analysis

Encoded PowerShell observed. Decoded content (data only): SYNTHETIC - test; execution context and intent require review.

Classification: suspicious; operation outcome: unknown

Evidence references: 098d0cfb012744a7a5c0fc785bda47b233f32165e4d3d8e809d05537d42a89e4

log_analysis

SI=2025-01-01T00:00:00+00:00 disagrees with FN=2026-10-13T03:10:00+00:00. Possible metadata manipulation; copying/restoration must be ruled out.

Classification: suspicious; operation outcome: unknown

Evidence references: f24f705cdd6d7eb9259b9b47ccbd92af6e8d58f10e6e763675428d7343cf01eb

log_analysis

Unsigned executable launched from ProgramData; source context={}

Classification: context; operation outcome: unknown

Evidence references: 49d1830b659fd7f78ec440497974cca6cd7d2245de23129649949858743f8e48

cloud_container

Service-account key request unknown for svc-backup@specula-lab-synthetic.iam.gserviceaccount.com; authorization and intent require review.

Classification: suspicious; operation outcome: unknown

Evidence references: 1eeb22b6dfc00d349c2a96388bb983da13d1dd328b73d3e14c6e62ebd54a64f2

cloud_container

storage.objects.list; outcome=unknown; principal=svc-backup@specula-lab-synthetic.iam.gserviceaccount.com; resource=projects/_/buckets/specula-sensitive-backups

Classification: context; operation outcome: unknown

Evidence references: 39004ff92ab6b04868ca57392ee9c50852e5c671ea0a507b7faa9ba69f0fdbc7

cloud_container

CLOUD_PRINCIPAL_ACTIVITY for source key svc-backup@specula-lab-synthetic.iam.gserviceaccount.com: [{"operation": "google.iam.admin.v1.CreateServiceAccountKey", "outcome": "unknown", "resource": "projects/-/serviceAccounts/svc-backup@specula-lab-synthetic.iam.gserviceaccount.com", "status": null, "time": "2026-10-13T03:20:00Z", "verdict": null}, {"operation": "storage.objects.list", "outcome": "unknown", "resource": "projects/_/buckets/specula-sensitive-backups", "status": null, "time": "2026-10-13T03:22:00Z", "verdict": null}].
Failed/unknown effects stay distinct; disagreements and authorization require review. No causal actor or family conclusion.

Classification: unresolved; operation outcome: unknown

Evidence references: 1eeb22b6dfc00d349c2a96388bb983da13d1dd328b73d3e14c6e62ebd54a64f2, 39004ff92ab6b04868ca57392ee9c50852e5c671ea0a507b7faa9ba69f0fdbc7

identity

Source reports OpenProcess on lsass.exe; completed extraction not established.

Classification: suspicious; operation outcome: unknown

Evidence references: ef180ee7f7b9b03e282ce8cdadc2f02041c35a1a06ae3aab2a40a898f70a5974

identity

Identity audit event 4672; protocol=None; encryption=None; special privileges or legacy encryption are review context, not proof of credential theft.

Classification: context; operation outcome: unknown

Evidence references: 84627e88414ba5159935eb8979a6f2f8542beea2c39dc765a8b40b716850be57

identity

2 failed logons followed by successful logon for administrator from 203.0.113.45. Possible credential compromise; count alone does not prove brute force.

Classification: suspicious; operation outcome: unknown

Evidence references: 7d0fe397f89de4360afd58e97dd9da02a9d526f77fe44db22232e2c4bc0205e0, 1890fa9e2c4b9d7700bf83f3568cf731f64e336459e0a8b9cdf541ff0b0a13bb, 99a5c5a73b882995178dcc5c9dc757d03c3e2cacc3d53ff142158bb5bfec7198

insider_threat

Source behavioral anomaly: remote_login; insider identity/intent not established.

Classification: suspicious; operation outcome: unknown

Evidence references: 9306da2738863b2503ec39fd2d814104fcaf96e857c18d4ddda967ff4928a1a8

insider_threat

Source behavioral anomaly: large_outbound_transfer; insider identity/intent not established.

Classification: suspicious; operation outcome: unknown

Evidence references: e29aaaf03ab7cb8d40ca54ad8a6966642fd2b1fe446564e84166f01f2bc132ad

malware_stylometry

Source reports unsigned executable: Unsigned executable launched from ProgramData; no malware-family attribution.

Classification: suspicious; operation outcome: unknown

Evidence references: 49d1830b659fd7f78ec440497974cca6cd7d2245de23129649949858743f8e48

malware_stylometry

Source reports unsigned executable: Unsigned executable launched from ProgramData; no malware-family attribution.

Classification: suspicious; operation outcome: unknown

Evidence references: 7798710c4e2200c1c09ce0a586d759092692b9d46e057922d0f78adeb10cf379

memory_forensics

Source-reported memory observation: Credential material pattern observed in memory snapshot; coverage=unknown

Classification: unresolved; operation outcome: unknown

Evidence references: 12c4270ebde02352981786b45f267276732f64fee9fc8de75cd8b86b0b215f06

memory_forensics

Source-reported memory observation: RWX private memory region with PE-like header; coverage=unknown

Classification: unresolved; operation outcome: unknown

Evidence references: d8f86bc22440cafcde1591af767b093bc7e90288cfbca010bf92055d4a093b64

supervisor_followup

Request additional analysis or examination of the executable to determine its nature and intent.; examination status=complete; Case-scoped recorded identifiers only; absence is not proof of non-occurrence. Source relationships do not establish causation or authorization.

Classification: context; operation outcome: unknown

Evidence references: 12c4270ebde02352981786b45f267276732f64fee9fc8de75cd8b86b0b215f06, 1890fa9e2c4b9d7700bf83f3568cf731f64e336459e0a8b9cdf541ff0b0a13bb, 450bb3a062a93585350e3e89e9ba6c53ff2dc75ffa3ca6908a732f9c53628bce, 49d1830b659fd7f78ec440497974cca6cd7d2245de23129649949858743f8e48, 7798710c4e2200c1c09ce0a586d759092692b9d46e057922d0f78adeb10cf379, 7d0fe397f89de4360afd58e97dd9da02a9d526f77fe44db22232e2c4bc0205e0, 84627e88414ba5159935eb8979a6f2f8542beea2c39dc765a8b40b716850be57, 9306da2738863b2503ec39fd2d814104fcaf96e857c18d4ddda967ff4928a1a8, 99a5c5a73b882995178dcc5c9dc757d03c3e2cacc3d53ff142158bb5bfec7198, bd5eef55cd225d45b87b64b1db021c03c64ad4ff4933a2c7c8b3a003230e8b52, d8f86bc22440cafcde1591af767b093bc7e90288cfbca010bf92055d4a093b64, e29aaaf03ab7cb8d40ca54ad8a6966642fd2b1fe446564e84166f01f2bc132ad, ef180ee7f7b9b03e282ce8cdadc2f02041c35a1a06ae3aab2a40a898f70a5974, fb5ca28e786e848f56cbc1cc10f319615ef7b0a1abe142d5245003ab0ca0e66e

supervisor_followup

Investigate the source IP (203.0.113.45) and verify if it is associated with known malicious activity. Check for any additional logs or network traffic related to the administrator user.; examination status=complete; Case-scoped recorded identifiers only; absence is not proof of non-occurrence. Source relationships do not establish causation or authorization.

Classification: context; operation outcome: unknown

Evidence references: 9306da2738863b2503ec39fd2d814104fcaf96e857c18d4ddda967ff4928a1a8

supervisor_followup

Investigate the destination IP (198.51.100.77) and verify if it is associated with known malicious activity. Check for any additional logs or network traffic related to the administrator user.; examination status=complete; Case-scoped recorded identifiers only; absence is not proof of non-occurrence. Source relationships do not establish causation or authorization.

Classification: context; operation outcome: unknown

Evidence references: e29aaaf03ab7cb8d40ca54ad8a6966642fd2b1fe446564e84166f01f2bc132ad

supervisor_followup

Investigate the process tree and memory dumps of svchost-helper.exe and lsass.exe for signs of credential extraction.; examination status=complete; Case-scoped recorded identifiers only; absence is not proof of non-occurrence. Source relationships do not establish causation or authorization.

Classification: context; operation outcome: unknown

Evidence references: ef180ee7f7b9b03e282ce8cdadc2f02041c35a1a06ae3aab2a40a898f70a5974

supervisor_followup

Investigate the source IP for any signs of automated attacks or repeated attempts to access the system.; examination status=complete; Case-scoped recorded identifiers only; absence is not proof of non-occurrence. Source relationships do not establish causation or authorization.

Classification: context; operation outcome: unknown

Evidence references: 1890fa9e2c4b9d7700bf83f3568cf731f64e336459e0a8b9cdf541ff0b0a13bb, 7d0fe397f89de4360afd58e97dd9da02a9d526f77fe44db22232e2c4bc0205e0, 99a5c5a73b882995178dcc5c9dc757d03c3e2cacc3d53ff142158bb5bfec7198

supervisor_followup

Review the privilege assignment logs and correlate them with other events to assess potential privilege escalation.; examination status=complete; Case-scoped recorded identifiers only; absence is not proof of non-occurrence. Source relationships do not establish causation or authorization.

Classification: context; operation outcome: unknown

Evidence references: 84627e88414ba5159935eb8979a6f2f8542beea2c39dc765a8b40b716850be57

supervisor_followup

Request additional analysis or examination of the executable to determine its nature and intent.; examination status=complete; Case-scoped recorded identifiers only; absence is not proof of non-occurrence. Source relationships do not establish causation or authorization.

Classification: context; operation outcome: unknown

Evidence references: 450bb3a062a93585350e3e89e9ba6c53ff2dc75ffa3ca6908a732f9c53628bce, 49d1830b659fd7f78ec440497974cca6cd7d2245de23129649949858743f8e48, 7798710c4e2200c1c09ce0a586d759092692b9d46e057922d0f78adeb10cf379, 79a8f04efc3dac3410ed4d430ec0132309ec0a744503aa36e3666811cdd3ba42

supervisor_followup

Verify the synthetic nature of the finding and cross-check with other memory artifacts or logs for potential credential leaks.; examination status=complete; Case-scoped recorded identifiers only; absence is not proof of non-occurrence. Source relationships do not establish causation or authorization.

Classification: context; operation outcome: unknown

Evidence references: 12c4270ebde02352981786b45f267276732f64fee9fc8de75cd8b86b0b215f06

supervisor_followup

Request additional analysis or examination of the executable to determine its nature and intent.; examination status=complete; Case-scoped recorded identifiers only; absence is not proof of non-occurrence. Source relationships do not establish causation or authorization.

Classification: context; operation outcome: unknown

Evidence references: 450bb3a062a93585350e3e89e9ba6c53ff2dc75ffa3ca6908a732f9c53628bce, 49d1830b659fd7f78ec440497974cca6cd7d2245de23129649949858743f8e48, 7798710c4e2200c1c09ce0a586d759092692b9d46e057922d0f78adeb10cf379, 79a8f04efc3dac3410ed4d430ec0132309ec0a744503aa36e3666811cdd3ba42

supervisor_followup

Perform a detailed analysis of the memory region using forensic tools to determine its origin and potential malicious intent.; examination status=complete; Case-scoped recorded identifiers only; absence is not proof of non-occurrence. Source relationships do not establish causation or authorization.

Classification: context; operation outcome: unknown

Evidence references: d8f86bc22440cafcde1591af767b093bc7e90288cfbca010bf92055d4a093b64

8. Event chronology and exhibit index

Exhibit hashes below identify normalized JSON, not original source-file bytes.

Exhibit / UID	Time (UTC)	OCSF class	Recorded observation	Normalized JSON SHA-256
E-00001 / f24f705cdd6d7eb9259b9b47ccbd92af6e8d58f10e6e763675428d7343cf01eb	2025-01-01T00:00:00+00:00	1001	Normalized event	74df6785088a3dd41f08adfb8252fea144dd3a796b1d3380eb3a0f11a28e440b
E-00002 / bd5eef55cd225d45b87b64b1db021c03c64ad4ff4933a2c7c8b3a003230e8b52	2026-10-13T03:00:00+00:00	3002	Normalized event	31d13a64dae7aa965d01378dc69cba3838535724a5ebb6ea7cb36571c54f1ed0
E-00003 / fb5ca28e786e848f56cbc1cc10f319615ef7b0a1abe142d5245003ab0ca0e66e	2026-10-13T03:00:00+00:00	2004	Normalized event	b490e1d02ecba89bb7c6d985563ca1e0457852317512db69d539b216eb327ec1
E-00004 / 7d0fe397f89de4360afd58e97dd9da02a9d526f77fe44db22232e2c4bc0205e0	2026-10-13T03:05:00+00:00	3002	Normalized event	eecba996d4b2f0728982f7ddc944f119190630a60477473054579f0c2cce31cc
E-00005 / 1890fa9e2c4b9d7700bf83f3568cf731f64e336459e0a8b9cdf541ff0b0a13bb	2026-10-13T03:06:00+00:00	3002	Normalized event	44588bed9f66032b0de705f049f680d42c7149f48629ff3511d2c7748da2addc
E-00006 / 9306da2738863b2503ec39fd2d814104fcaf96e857c18d4ddda967ff4928a1a8	2026-10-13T03:07:00+00:00	2004	Normalized event	cee80029a4cdf9796e5812bc9983431ecba7c2e5420cbdd715ead5a8965efdcb
E-00007 / 99a5c5a73b882995178dcc5c9dc757d03c3e2cacc3d53ff142158bb5bfec7198	2026-10-13T03:07:00+00:00	3002	Normalized event	3b34b05038003230a9f079c133cb13fa09b35e84bd891783d3d2acd43b59fa4d
E-00008 / 098d0cfb012744a7a5c0fc785bda47b233f32165e4d3d8e809d05537d42a89e4	2026-10-13T03:08:00+00:00	1007	C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe	d2be8833e3bb6a8f9685d078552fedf5631cf050b5c1c12860998c583ef383dc

Exhibit / UID	Time (UTC)	OCSF class	Recorded observation	Normalized JSON SHA-256
E-00009 / 450bb3a062a93585350e3e89e9ba6c53ff2dc75ffa3ca6908a732f9c53628bce	2026-10-13T03:10:00+00:00	1001	Normalized event	6521974ccca736021fd445eb4fd3ece23ebe569987f6efc248d103afb4c5b43c
E-00010 / 84627e88414ba5159935eb8979a6f2f8542beea2c39dc765a8b40b716850be57	2026-10-13T03:11:00+00:00	3002	Normalized event	43dbf87770ded3ef231ee22679c78985313ddb9e9df61cff006e87a3cb92f731
E-00011 / 49d1830b659fd7f78ec440497974cca6cd7d2245de23129649949858743f8e48	2026-10-13T03:12:00+00:00	2004	Normalized event	4d4d3323cc718d0fe68062444db21763c2d7cc08f7f330316292 7b8983cd31e8
E-00012 / 7798710c4e2200c1c09ce0a586d759092692b9d46e057922d0f78adeb10cf379	2026-10-13T03:12:00+00:00	1007	svchost-helper.exe	f5f1bb8edf9c6cb0aa455c2b0220abfb1adc044b447b0f261bed654483b4d36d
E-00013 / 79a8f04efc3dac3410ed4d430ec0132309ec0a744503aa36e3666811cdd3ba42	2026-10-13T03:12:00+00:00	1007	C:\ProgramData\svchost-helper.exe	763dfd850369ec24d913cea38185ff279aa4bd5c633ac9e6ab46c502a2a91209
E-00014 / 12c4270ebde02352981786b45f267276732f64fee9fc8de75cd8b86b0b215f06	2026-10-13T03:13:00+00:00	2004	Normalized event	0c789129a984d99163009a45f95273ceade921aa47597dedf715e5f963a7e75e
E-00015 / d8f86bc22440cafcde1591af767b093bc7e90288cfbca010bf92055d4a093b64	2026-10-13T03:13:00+00:00	2004	Normalized event	c2b5ff773b005b430a1aaf9fe3f5974cb534e8ceec52ccf269e824a0caeffbd8
E-00016 / ef180ee7f7b9b03e282ce8cdadc2f02041c35a1a06ae3aab2a40a898f70a5974	2026-10-13T03:13:00+00:00	2004	Normalized event	b54e2372d7fc287c534ba129f62c096adbcdd16154651a4d0f1864d864a4a098

Exhibit / UID	Time (UTC)	OCSF class	Recorded observation	Normalized JSON SHA-256
E-00017 / e29aaaf03ab7cb8d40ca54ad8a6966642fd2b1fe446564e84166f01f2bc132ad	2026-10-13T03:14:00+00:00	2004	Normalized event	05a9427d74848e1f04a9a165c76d8835ef2f081585aba0b230167d00da4a84ba
E-00018 / c0d8a9cb4a8f9cb43fba1fc83bd2fadb9c86afa6d16a15b400dc242185c71ba7	2026-10-13T03:15:00+00:00	1007	C:\Windows\System32\cmd.exe	12a9b1fe87361d7e9db1dcf136e3b550476c57f0bb45fe3cb5739eb4ee7e097f
E-00019 / 1eeb22b6dfc00d349c2a96388bb983da13d1dd328b73d3e14c6e62ebd54a64f2	2026-10-13T03:20:00+00:00	6003	Normalized event	5b79977c88ef137ec3e8f815e99c321f9ef034a051fa335947972464b309e2ed
E-00020 / 39004ff92ab6b04868ca57392ee9c50852e5c671ea0a507b7faa9ba69f0fdbc7	2026-10-13T03:22:00+00:00	6003	Normalized event	fd85a0be6a90857a71928d56af5a9279ef26544ad3379bfccbcfd36856cb6a7f

9. Limitations and alternative explanations


```
[
  "actor_sequence_inferred",
  "actor_sequence_unavailable",
  "additional_evidence_requested",
  "agent_review_incomplete",
  "model_review_coverage_incomplete",
  "profile_ranking_ambiguous",
  "security_screening_degraded",
  "semantic_mapping_unavailable",
  "source_clock_skew_unverified",
  "unsupported_technique_id",
  "uploaded_record_coverage_limited"
]
```


Unsupported records, source selection, excluded ports and missing telemetry can reduce coverage. Model inference, attribution similarity and event ordering do not establish actor identity, intent or causality. Collection integrity and error rates require independent validation.

10. Examiner and independent reviewer approval

Examiner name / qualifications: _____

Scope and methods reviewed: _____

Examiner signature / date: _____

Independent reviewer signature / date: _____

Approval status: UNSIGNED. No signature or certification is generated by this system.

Specula examination template v1 • Case LOCAL-2f783c8d4d5941c2a6ad4442db1a6729 • End of report